MILESTONE 3 · MODEL ARCHITECTURE

ContextGuard

Finding and repairing cell-level errors in tables

A five-stage pipeline, from the dirty table to a calibrated repair

5 stages	**3,500** cells run end to end
34.9M trainable LoRA parameters (0.44%)	**93** tests pass

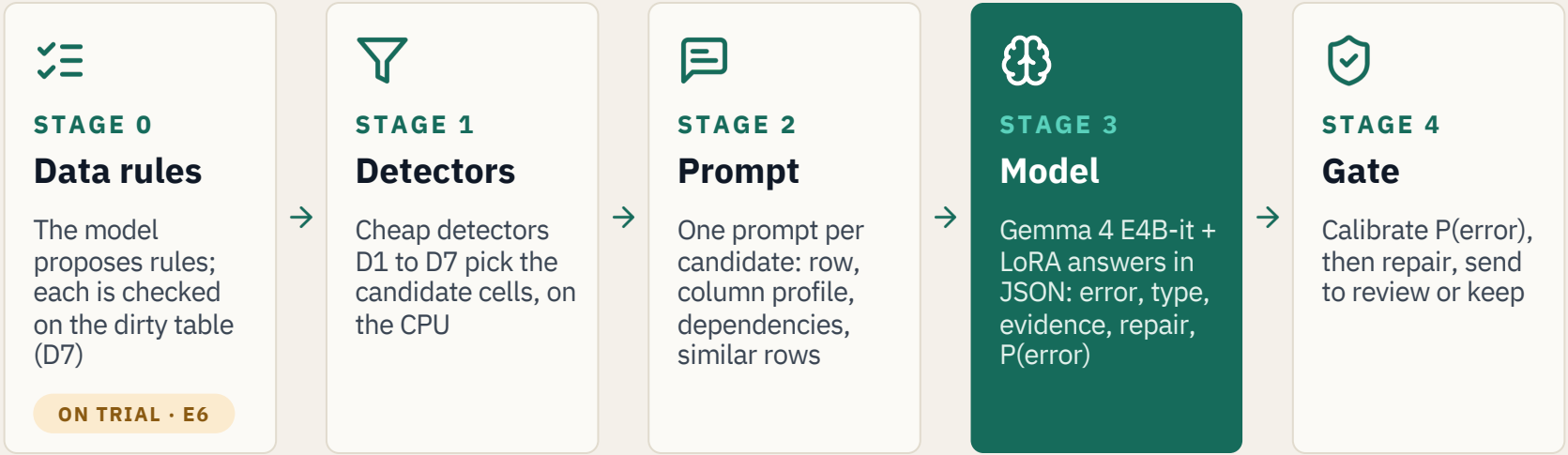


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DSAI Lab Project, IIT Madras BS · Group 1 · 5 November 2026

PIPELINE · REPORT §2, FIGURE 1

Five stages from table to repair



The clean table is never read by the pipeline

Runs locally; one model serves stages 0 and 3

Output: cleaned table, change log, review list

MODEL · REPORT §4.2, §5.1, §7.1

The model: Gemma 4 E4B-it with LoRA

LORA ADAPTER

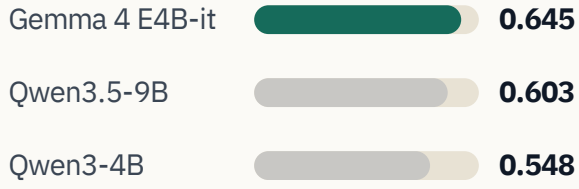
34.9M

trainable parameters, 0.44% of the model

- ✓ Rank 16, alpha 32
- ✓ q, k, v, o, gate, up, down
- ✓ 258 adapted matrices
- ✓ Base weights frozen

42	2,560	8 / 2	262k
decoder layers	hidden size	query / key-value heads	vocabulary

WHY THIS MODEL · M1 SCORE



Also the lowest false-repair rate and 100% valid JSON

TRAINING SETUP

- AdamW, cosine schedule
- Learning rate 2e-4
- Batch 32
- Early stopping
- Loss on the answer only

DATA · REPORT §8.1, §8.2

Dataset organisation and the M3 subset

data/raw
pinned downloads

→

data/processed
cell files, 73,979 examples

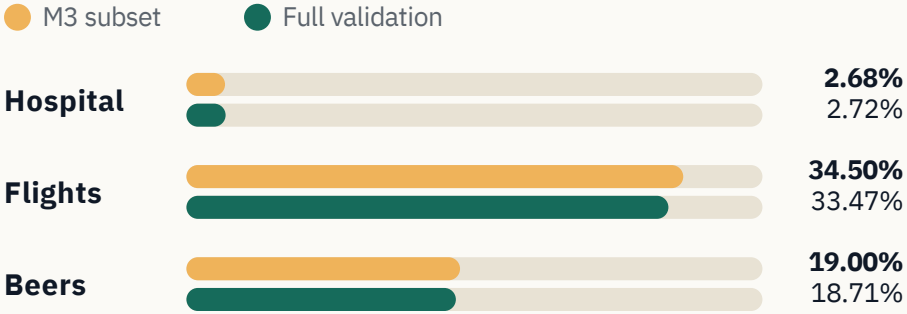
→

data/splits
train, val, test per table (entity split)

→

data/subsets/m3
100 validation rows each of Hospital, Flights, Beers

ERROR RATE · SUBSET VS FULL VALIDATION



3,500
cells, every cell kept

448
errors

- ✓ No test row is used
- ✓ Only a 20-record sample in git; one command rebuilds the rest

The training data loader

CHAT TEMPLATE, FROM THE TOKENIZER

```
<bos><|turn>user
  … prompt …
<turn|>
<|turn>model
  … JSON answer …
```

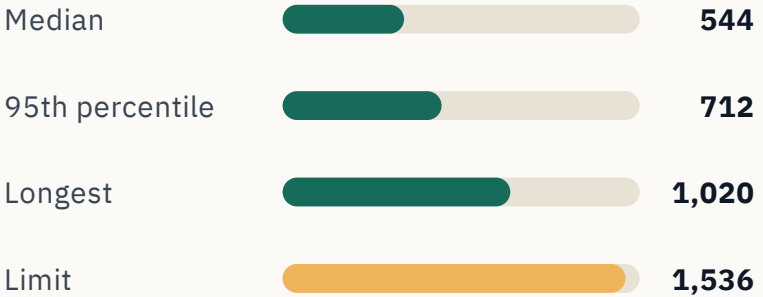
Same text in training and inference, checked by a test

LABELS · WHAT THE LOSS SEES

prompt · −100	**answer · loss**	pad

`int64 tensors: input_ids, attention_mask, labels`

TOKEN LENGTHS · GEMMA 4 TOKENIZER



0 training examples over the 1,536-token limit, so nothing is cut

DETECTORS · REPORT §4.3, §12

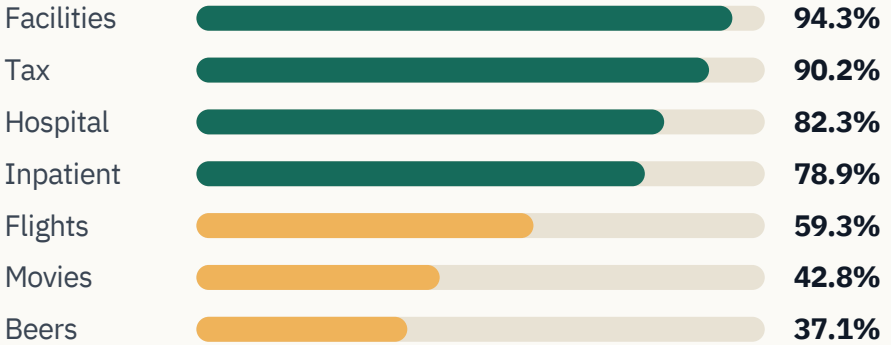
Stage 1: cheap detectors and candidate recall

SIX DETECTORS, ONE INTERFACE

D1 Missing value	**D2** Rare pattern
D3 Robust z (MAD)	**D4** Isolation Forest
D5 Dependencies (g3)	**D6** Edit distance

```
detect(dirty_df)
```

CANDIDATE RECALL · VALIDATION SPLITS



7% to 51%
of the cells pass the filter, depending on the table

7.9 s
per 1,000 rows on the CPU, a few MB of memory

Weak spot
Wrong values that look normal: Beers ounces, Flights times

BASELINES · REPORT §5.4

Baselines: first runs on the test splits

Detection F1	Hospital	Flights	Beers	Rayyan	Movies
B1 · constraint check	0.13	0.35	0.08	0.00	0.00
B2 · dBoost	0.09	0.57	0.53	0.11	0.00
B3 · Raha (reduced)	0.41	**0.88**	**0.95**	0.22	0.12

Rule D8: the 20 labelled tuples are excluded from scoring. Darker cell = higher F1.

WHAT THE PATTERN SAYS

Rules work where dependencies exist, statistics over-flag, and Raha is strong with labels on clean, structured tables but weak on Rayyan and Movies.

WHAT THESE RUNS ARE

B1 checks violations of the mined constraints; HoloClean not yet run

B2 is dBoost with one setting; the grid comes later

B3 is Raha with a reduced strategy set; Baran not yet run

They test the baseline files and the scoring; full runs with repair come in M4 and M5.

TRAINING SIGNAL · REPORT §7.2, §7.3

Loss, confidence and calibration

1 · LOSS

Cross-entropy over the answer tokens; prompt and padding masked. No class weights: training data is balanced.

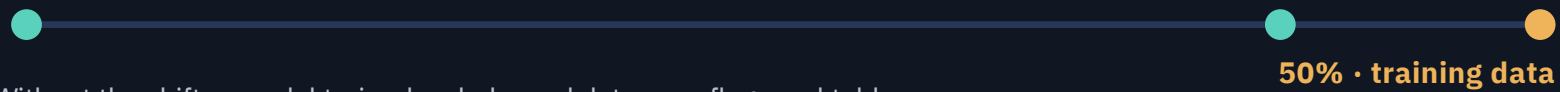
2 · CONFIDENCE

P(error) is the probability of "true" against "false" at the first answer token, after {"is_error":

3 · CALIBRATION

Temperature scaling on validation data, then a shift by the table's real error rate.

WHY CALIBRATION: REAL ERROR RATES ARE FAR BELOW 50%



Without the shift, a model trained on balanced data over-flags real tables

EVALUATION · REPORT §7.4, §8.3, §10

Evaluation metrics and output guardrails

Detection

Precision, recall, F1; error-type macro-F1

Repair

Repair accuracy; false-repair rate

Safety

False alarms per 1,000 correct cells; calibration error

Coverage

Candidate recall; PR-AUC; bootstrap intervals

HOW WE SCORE

Per table and per error type, with the number of errors behind each entry; every cell at its natural error rate; format-only cells left out. No single overall score.

OUTPUT GUARDRAILS

- Strict JSON schema check on every answer
- Contradictions are rejected, such as "no error" with a repair
- An invalid answer is never repaired: it goes to review

END TO END · REPORT §3.6

The whole pipeline on the subset

0.57
F1

0.70
Precision

0.48
Recall

21.9%
of cells are candidates; they hold 58.9% of the errors

F1 PER TABLE · ZERO-SHOT MODEL



3,500 cells, every answer valid JSON; about 0.30 s per candidate cell

REPAIRED CORRECTLY · HOSPITAL PHONE

3342xx2100 → 3342882100

Found, typed as a pattern error, repaired with P 1.00

A CORRECT VALUE FLAGGED · BEERS ABV

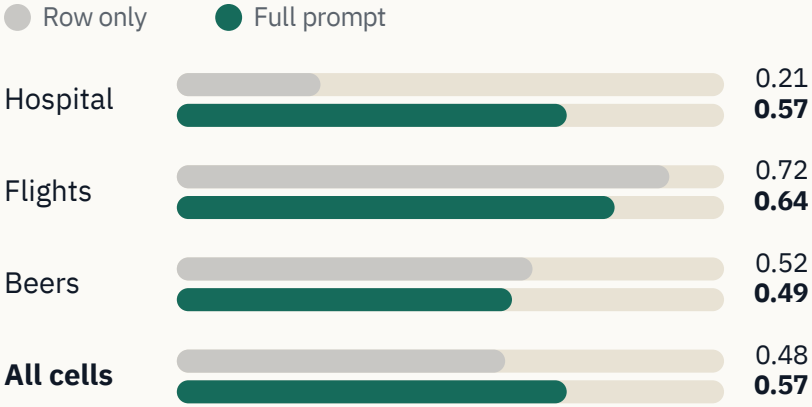
0.063 → 0.063

"Repaired" to itself: a post-processing rule for M4

EXPERIMENTS · REPORT §3.6, §7.1

What the first experiments show

E2 · CONTEXT IN THE PROMPT · F1, ZERO-SHOT



False alarms 86 → 30 per 1,000 correct cells. Prompt unchanged for M4; E2 is repeated on the fine-tuned model.

STAGE 0 TRIAL

19 of 27 rules kept, 0 errors added

Not fine-tuned: only format and range rules in M4; kept if E6 shows a gain

LORA SMOKE RUN

Loss 1.22 → 0.08

100 steps; adapter saved and reloaded with the same loss

WITH THE 100-STEP ADAPTER

Repair 14% → 28%

PR-AUC 0.80 → 0.87, but more false alarms: calibration needed

NEXT STEPS · REPORT §14, §16

Risks and the plan for Milestone 4

MAIN RISKS

- ⚠ Candidate recall caps the system
- ⚠ A confidently wrong model
- ⚠ Outliers learned only from synthetic data
- ⚠ Heuristic error types in the external tables

MILESTONE 4 · 19 NOVEMBER

- ✓ Train on all 55,091 examples, early stopping
- ✓ Tune hyper-parameters over several seeds
- ✓ Fit calibration and the gate on validation data
- ✓ Real HoloClean, Raha with Baran, dBoost grid

EXPERIMENTS IN M4

E1
Base against fine-tuned

E2
Context, on the fine-tuned model

E5
With and without synthetic data

E6
With and without stage 0 (D7)